Sucheol Woo

Manufacturing & Process Engineering | Prepreg Layup & Autoclave · 5-Axis CNC · CATIA V5 · Test & Validation · FAA A&P
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EDUCATION

Purdue University, MS, Aviation and Aerospace Management · GPA 3.62/4.00 — Expected May 2027
Purdue University, BS, Aeronautical Engineering Technology — May 2025
Idaho State University, AAS, Aircraft Maintenance Technology — Jun 2020

EXPERIENCE

Graduate Research Assistant, Purdue Composites Manufacturing & Simulation Center (CMSC) — Aug 2025 – Present

CFRP hot-water and freeze–thaw testing

- Fabricated carbon/epoxy test panels by prepreg layup, vacuum bagging and autoclave cure
- Planned and ran a 114-day hot/wet and freeze–thaw conditioning program
- Tested conditioned specimens per ASTM D2344, D7264 and D7028 (DMA) and inspected failures

FAA helicopter fuel-tank drop tests

- Designed sensor-protection fixtures in CATIA V5 and machined them from aluminum (Haas VF-3, UMC-500)
- Wired and soldered accelerometer and pressure-sensor leads and set up the DAQ; captured every Phase 2 drop
- Measured post-drop tank deformation with a FARO Arm 3D scanner

Large composite part machining · Thermwood 5-axis router

- Fixtured, 3D-scanned, set work offsets, verified NC programs and machined LSAM-printed parts up to 1.5 m
- Machined, assembled and vacuum leak-tested a 4-section wind turbine blade mold

CNC process monitoring

- Designed, wired and built a controller-independent Raspberry Pi module logging vibration, sound and temperature
- Validated the module during live cuts on a Thermwood 5-axis router

Undergraduate Research & Teaching Assistant, Purdue CMSC — Jan 2024 – May 2025

- Ran a 16-condition speed × feed DOE on 3D-printed CF-ABS tooling (Haas VF-2); found feed drives surface finish and spindle speed drives heat
- Built a carbon-fiber Purdue Pete mascot head by prepreg layup, vacuum bagging and autoclave cure
- Taught CATIA V5, 3DEXPERIENCE and G-code to 20+ students and supervised safe operation of Haas CNC mills

CH-47 Flight Crew, Republic of Korea Army — Apr 2021 – Oct 2022

- Inspected and maintained CH-47 helicopters, identifying mechanical and electrical discrepancies

CERTIFICATIONS, AWARDS & PUBLICATIONS

- FAA Airframe & Powerplant (A&P) Mechanic Certificate, Lean Six Sigma Green Belt (CSSC)
- 3rd Place, 2025 SAMPE Student Poster Contest (CNC machining study), Dean's List, Purdue and Idaho State
- Co-author, "IMPACT: Industrial Machine Perception via Acoustic Cognitive Transformer," arXiv:2507.06481 (2025)

SKILLS

Composites: prepreg layup, Gerber ply cutter, vacuum bagging, autoclave cure, waterjet, hot/wet conditioning
CAD/CAM & CNC: CATIA V5, 3DEXPERIENCE, NX, Mastercam, G-code; Haas VF-2/VF-3/UMC-500, Thermwood 5-axis
Electrical: wiring and soldering, sensor and DAQ setup, Raspberry Pi, electrical troubleshooting (FAA A&P)
Quality & process: ASTM testing, FARO Arm inspection, DOE, process validation, root cause analysis, Python